

Track 2 Component Offset Identification Diagnostic

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

Measured component-offset diagnostic over 969 CSV files, 1938 directional curves, and harmonic orders 0, 1, 3, 39, 40, 78, 81, 156, 162, 240.

Main Findings

Finding	Interpretation
Harmonic 0 mean absolute amplitude is 0.039935094 deg ; harmonic 1 is 0.017149350 deg .	The measured curve mean / a_0 proxy is the largest average component in the prepared diagnostic set.
Largest non-zero maximum amplitude is harmonic 156 at 0.034888028 deg .	a_0 is not the only component that can show large individual cases; high-order outliers still need inspection.
The current repository CSV grid has one curve per direction / speed / torque / temperature condition.	Repeatability cannot be estimated from the current canonical CSV set alone; external repeated-measurement data are required for a real repeatability conclusion.
Forward h0 values are consistently negative while backward values are mostly positive in the temperature-direction summaries.	Direction must remain a first-class diagnostic axis; a global-only offset correction would hide sign structure.

Harmonic Summary

Harmonic	Mean Abs Amp [deg]	P95 Abs Amp [deg]	Max Abs Amp [deg]
0	0.039935094	0.083114220	0.096616939
1	0.017149350	0.017278498	0.017859552
3	0.000875204	0.001134171	0.001460335
39	0.000790242	0.001588147	0.001878775
40	0.000566066	0.000896364	0.001100963
78	0.001124873	0.002319621	0.003014301
81	0.000222494	0.000411677	0.000610119
156	0.001042352	0.004207800	0.034888028
162	0.000876746	0.003036829	0.017653806
240	0.000661320	0.001756488	0.010896642

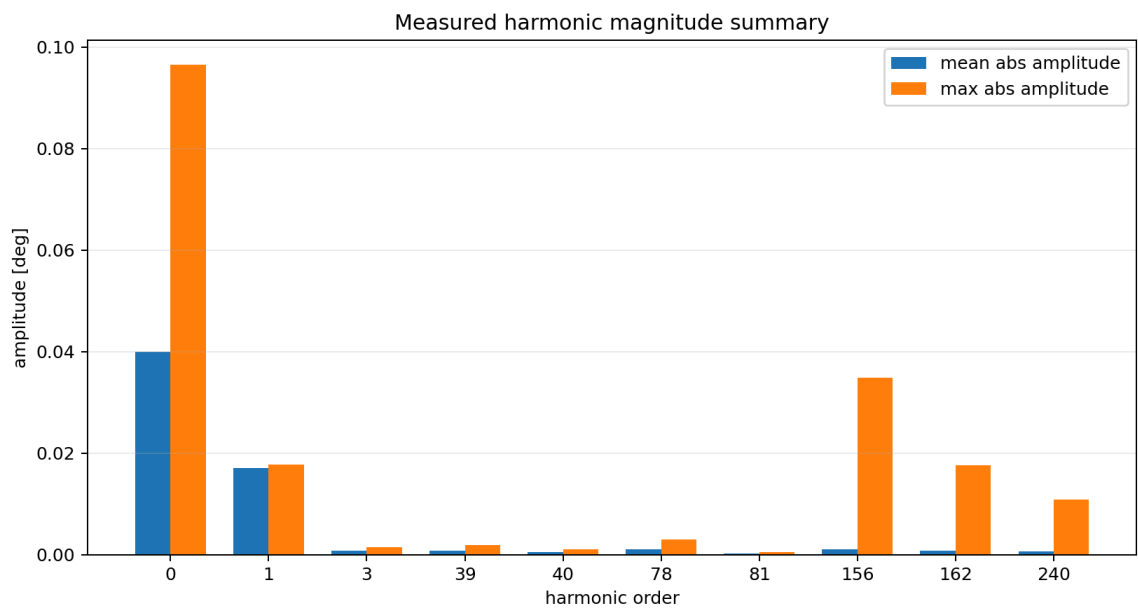
Temperature And Direction Summary

Direction	Temp	Curves	Mean h0	Std h0	Min h0	Max h0
backward	25	323	0.018656	0.020290	-0.029064	0.058645
backward	30	323	0.020811	0.020988	-0.020349	0.061768
backward	35	323	0.022132	0.021044	-0.018715	0.062035
forward	25	323	-0.055359	0.020531	-0.095686	-0.002487
forward	30	323	-0.056172	0.019818	-0.095808	-0.019389
forward	35	323	-0.055833	0.020146	-0.096617	-0.022198

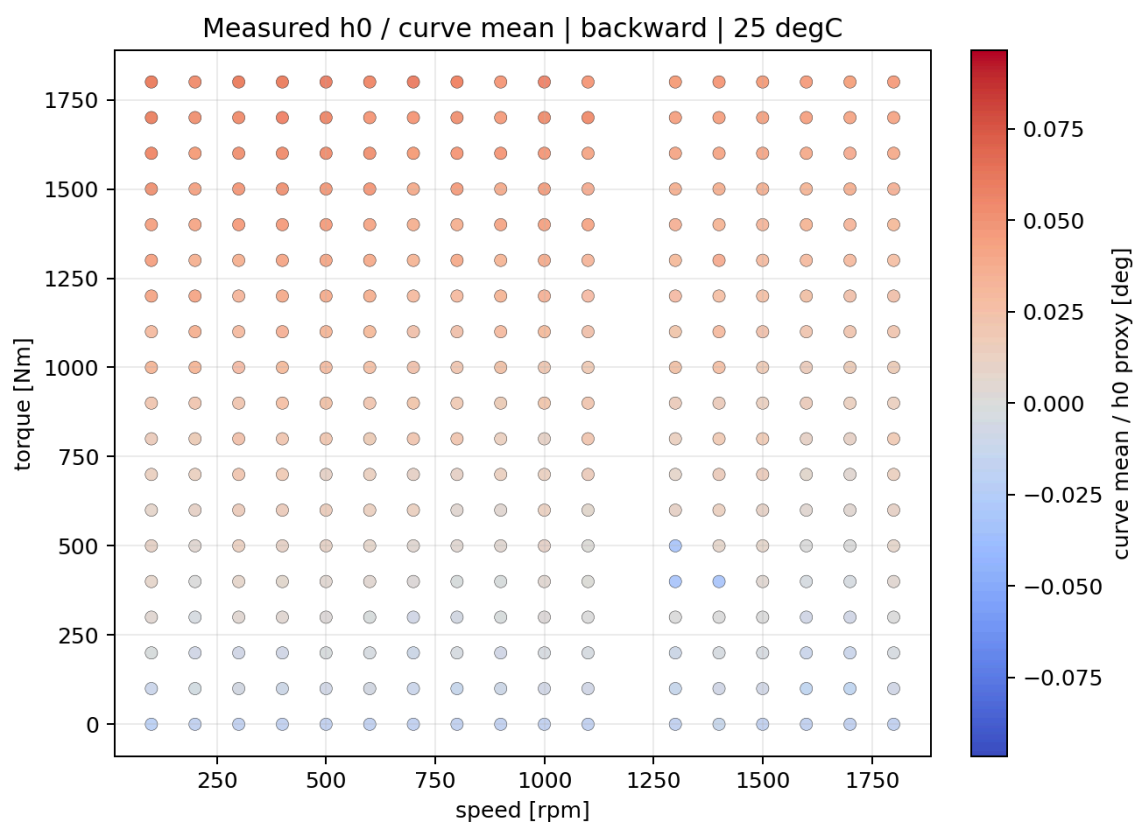
Largest Absolute h0 Cases

Rank	Direction	Speed	Torque	Temp	h0	P2P	File
1	forward	200	1800	35	-0.096617	0.048101	200.0rpm1800.0Nm35.0deg.csv
2	forward	300	1800	30	-0.095808	0.046346	300.0rpm1800.0Nm30.0deg.csv
3	forward	200	1800	25	-0.095686	0.047207	200.0rpm1800.0Nm25.0deg.csv
4	forward	700	1800	35	-0.094272	0.047849	700.0rpm1800.0Nm35.0deg.csv
5	forward	800	1800	30	-0.093640	0.046183	800.0rpm1800.0Nm30.0deg.csv
6	forward	500	1800	30	-0.093112	0.046271	500.0rpm1800.0Nm30.0deg.csv
7	forward	1000	1800	35	-0.092984	0.045909	1000.0rpm1800.0Nm35.0deg.csv
8	forward	1400	1800	35	-0.092720	0.048646	1400.0rpm1800.0Nm35.0deg.csv
9	forward	1000	1800	30	-0.092215	0.047390	1000.0rpm1800.0Nm30.0deg.csv
10	forward	300	1700	30	-0.091875	0.046462	300.0rpm1700.0Nm30.0deg.csv

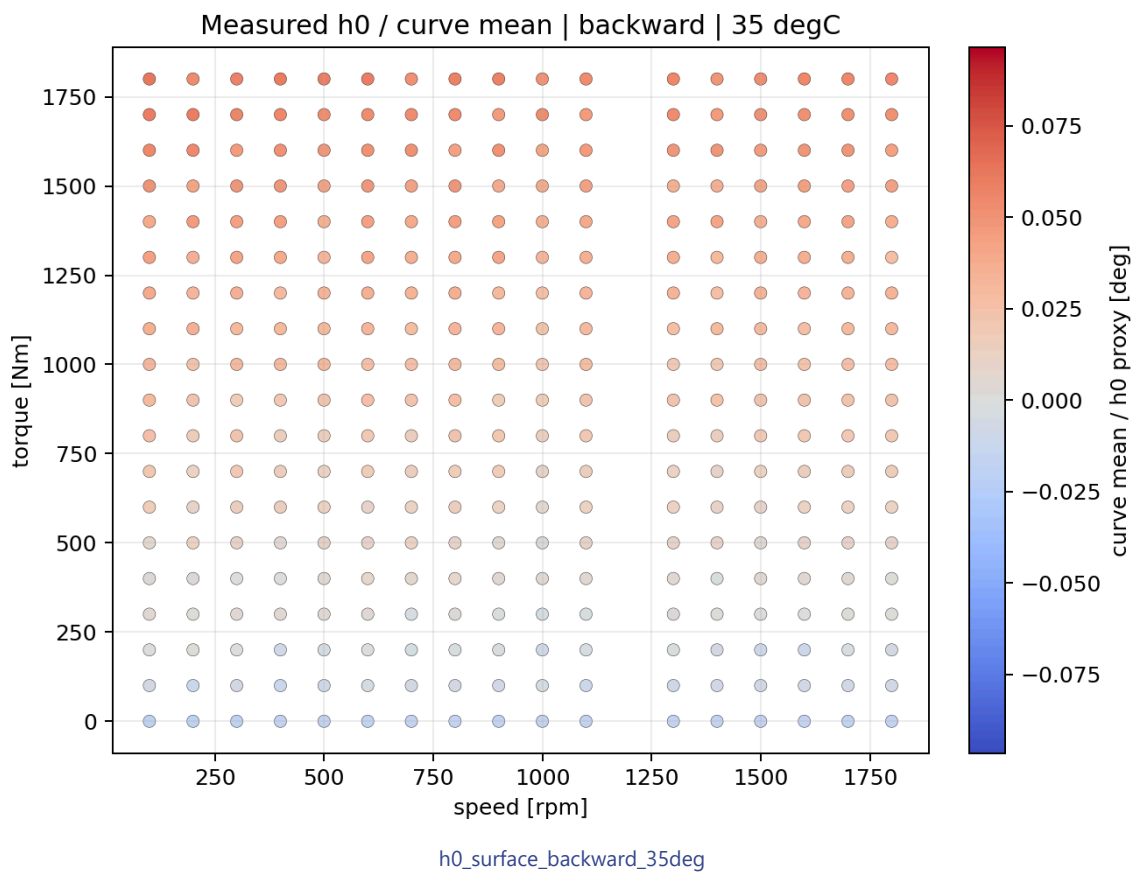
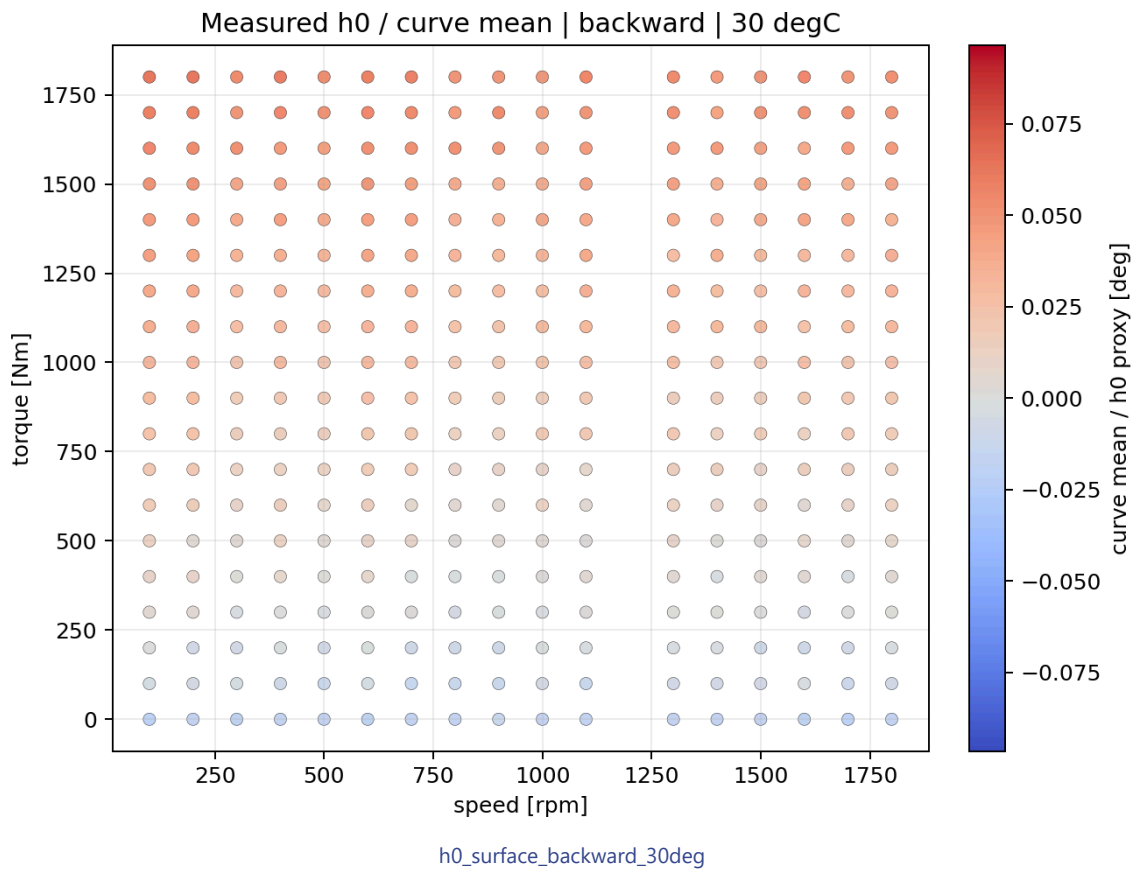
Figures

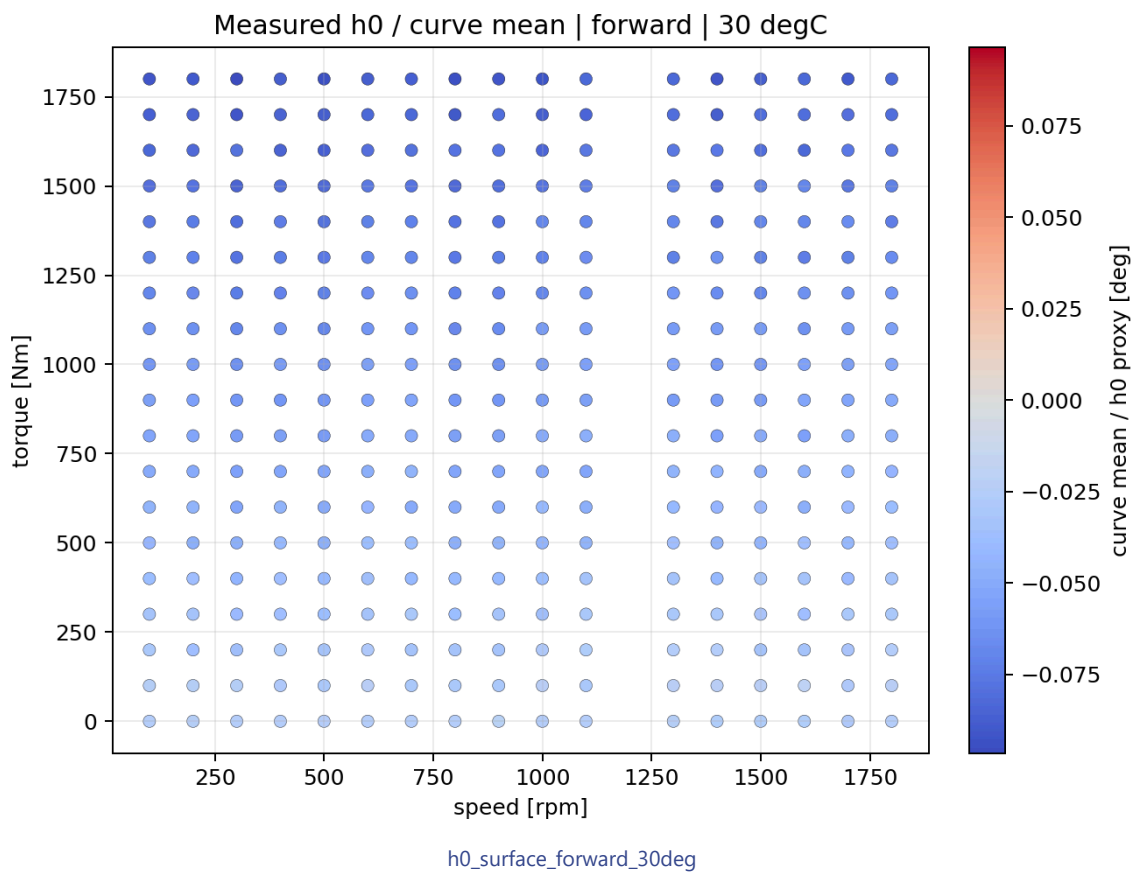
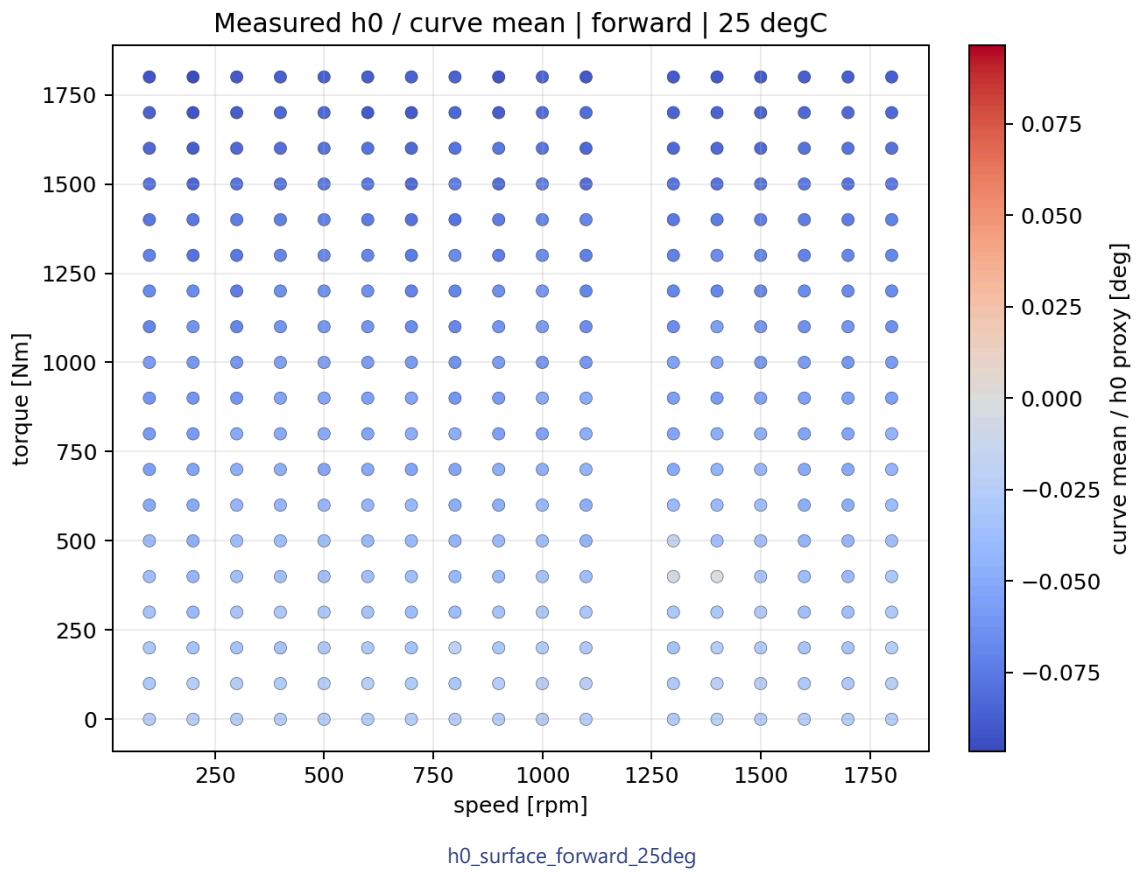


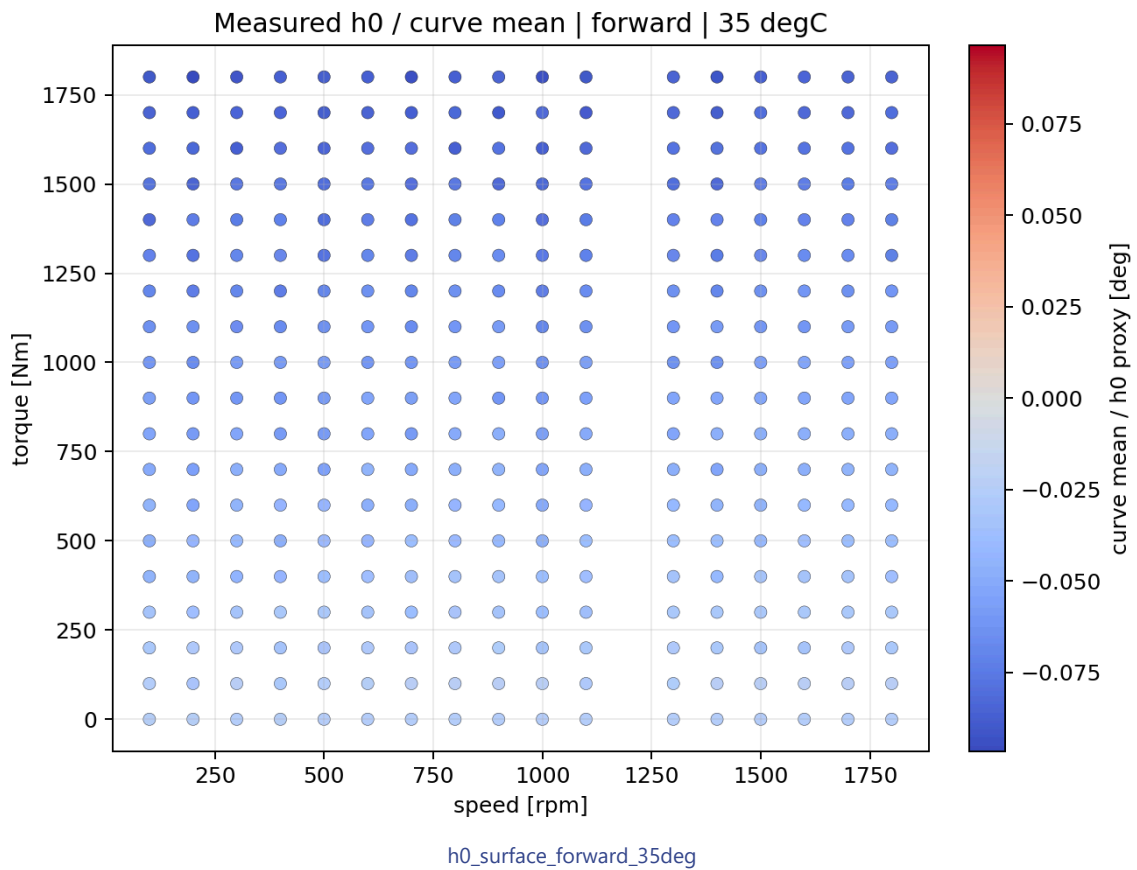
Harmonic magnitude summary



h0_surface_backward_25deg







Decision

a_0 / harmonic zero should stay the priority suspect because it is the largest average measured component and shows strong direction structure. It should not yet be documented as the sole confirmed cause of the Track 2 model offset. The next analysis should compare these measured h0 surfaces with Track 2D signed model-offset rows and inspect high-order outliers, especially where harmonic 156 , 162 , or 240 amplitudes spike.